

FENTRESS NALF, VA, USA

WMO: 725172

Lat: 36.695N

Lon: 76.136W

Elev: 16

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 13762

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.5	25.5	0.9	5.7	27.5	5.3	7.2	28.8	20.3	49.0	18.6	46.7	5.3	340	0.401

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	96.5	77.4	94.0	76.6	91.9	75.6	80.1	92.0	78.9	89.7	77.8	88.0	6.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.8	139.7	86.0	75.8	134.9	84.8	74.9	130.8	84.0	43.4	91.6	42.2	89.9	41.1	88.1	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.7	16.5	14.1	DB	14.1	100.1	7.2	1.6	9.0	101.2	4.8	102.1	0.8	103.0	-4.4	104.2
			WB	11.3	81.7	5.7	2.3	7.2	83.3	3.9	84.7	0.7	85.9	-3.4	87.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.2	42.6	48.0	51.7	63.7	71.4	78.4	81.3	79.6	75.2	65.2	53.0	47.7
	DBStd	15.79	10.20	10.92	9.61	10.09	7.58	5.58	4.83	4.03	5.58	8.45	8.48	10.12
	HDD50	757	273	154	96	6	0	0	0	0	0	3	59	166
	HDD65	2801	695	485	426	143	30	1	0	0	2	104	372	543
	CDD50	5585	43	99	150	416	662	852	970	916	757	475	149	96
	CDD65	2154	1	10	14	103	228	403	505	451	308	111	12	8
	CDH74	21159	2	118	152	1198	2045	4156	5649	4368	2558	815	55	43
	CDH80	8713	0	28	38	550	786	1807	2626	1771	874	223	6	4
Wind	WSAvg	5.5	5.9	6.4	7.2	7.1	5.7	4.9	4.4	4.0	5.1	5.1	5.3	5.3
Precipitation	PrecAvg	48.5	3.3	3.0	3.7	3.3	3.6	4.4	5.7	5.4	5.4	4.0	3.2	3.4
	PrecMax	62.6	6.3	7.8	9.2	6.5	5.4	11.3	9.6	12.8	15.2	8.3	10.7	7.4
	PrecMin	30.7	1.1	0.9	0.7	0.7	1.4	1.7	1.5	1.7	0.6	0.1	0.2	1.5
	PrecStd	7.8	1.3	1.5	1.6	1.4	1.2	2.0	2.0	2.7	3.6	2.1	2.3	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.5	84.8	84.0	101.9	96.0	98.1	99.0	96.0	93.8	90.7	79.6	79.1
		MCWB	63.9	62.4	64.1	74.2	74.0	77.8	80.0	78.5	77.4	73.5	68.3	69.8
	2%	DB	68.7	77.3	78.0	91.7	91.1	94.9	96.4	93.4	91.1	85.4	74.8	73.5
		MCWB	59.7	62.4	60.9	70.3	72.0	76.2	79.2	77.5	75.6	71.5	64.9	65.8
	5%	DB	64.3	72.2	73.2	86.0	88.0	92.3	94.0	91.2	88.3	82.0	71.2	69.1
		MCWB	56.1	59.2	59.3	66.9	71.7	75.4	77.7	76.3	74.2	70.1	63.0	63.0
	10%	DB	59.4	67.2	69.1	80.5	84.6	89.4	91.4	88.6	85.2	78.5	67.4	64.8
		MCWB	52.3	58.3	58.5	64.8	69.9	74.0	76.6	75.2	73.1	68.6	59.4	59.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.2	67.9	67.4	75.4	76.4	79.7	82.2	80.8	79.5	77.4	71.0	70.8
		MCDB	71.9	77.1	78.7	93.1	87.0	91.8	96.0	92.0	88.8	85.6	76.1	77.0
	2%	WB	61.7	65.3	64.7	72.7	74.8	78.2	80.6	79.4	78.2	74.9	67.5	67.0
		MCDB	67.5	73.4	74.3	89.5	85.3	90.2	93.6	90.3	87.2	82.1	73.5	72.4
	5%	WB	57.5	62.7	62.1	69.6	73.4	77.2	79.4	78.2	76.9	72.6	64.4	63.9
		MCDB	63.1	71.1	71.5	80.4	84.2	89.0	91.2	87.6	85.2	79.4	69.8	68.8
	10%	WB	52.5	58.6	58.4	66.7	72.1	76.1	78.1	77.0	75.4	70.2	60.6	59.8
		MCDB	58.8	66.3	67.5	77.9	81.9	86.6	88.7	85.3	82.1	76.9	66.9	65.1
Mean Daily Temperature Range	5% DB	MDBR	19.3	21.1	20.7	21.9	19.4	18.9	18.3	17.5	17.3	20.0	19.8	18.2
		MCDBR	26.4	28.6	27.8	28.2	22.7	20.7	20.1	19.8	20.6	21.5	22.0	19.7
		MCWBR	19.9	16.8	15.5	12.2	9.4	7.7	7.3	7.5	8.8	10.2	14.6	15.3
	5% WB	MCDBR	22.6	26.5	23.7	22.1	19.2	18.7	18.6	17.7	16.4	16.8	18.5	18.8
	MCWBR	19.9	17.9	16.4	12.3	8.7	7.7	7.3	7.4	7.7	8.9	14.6	15.8	
Clear-Sky Solar Irradiance	taub	0.311	0.323	0.349	0.391	0.435	0.484	0.509	0.502	0.407	0.358	0.327	0.316	
	taud	2.536	2.497	2.453	2.349	2.256	2.173	2.114	2.122	2.382	2.508	2.544	2.553	
	Ebn at Noon	281	291	291	283	270	256	249	247	268	274	273	272	
	Edh at Noon	25	29	33	39	43	47	50	48	35	29	24	23	
All-Sky Solar Radiation	RadAvg	745	988	1322	1689	1899	2029	1931	1741	1424	1133	843	666	
	RadStd	60	103	102	121	155	126	94	105	109	125	66	52	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (51 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+302	+194

Nomenclature: See separate page